

PKAF LEV

System Design & Analysis Document

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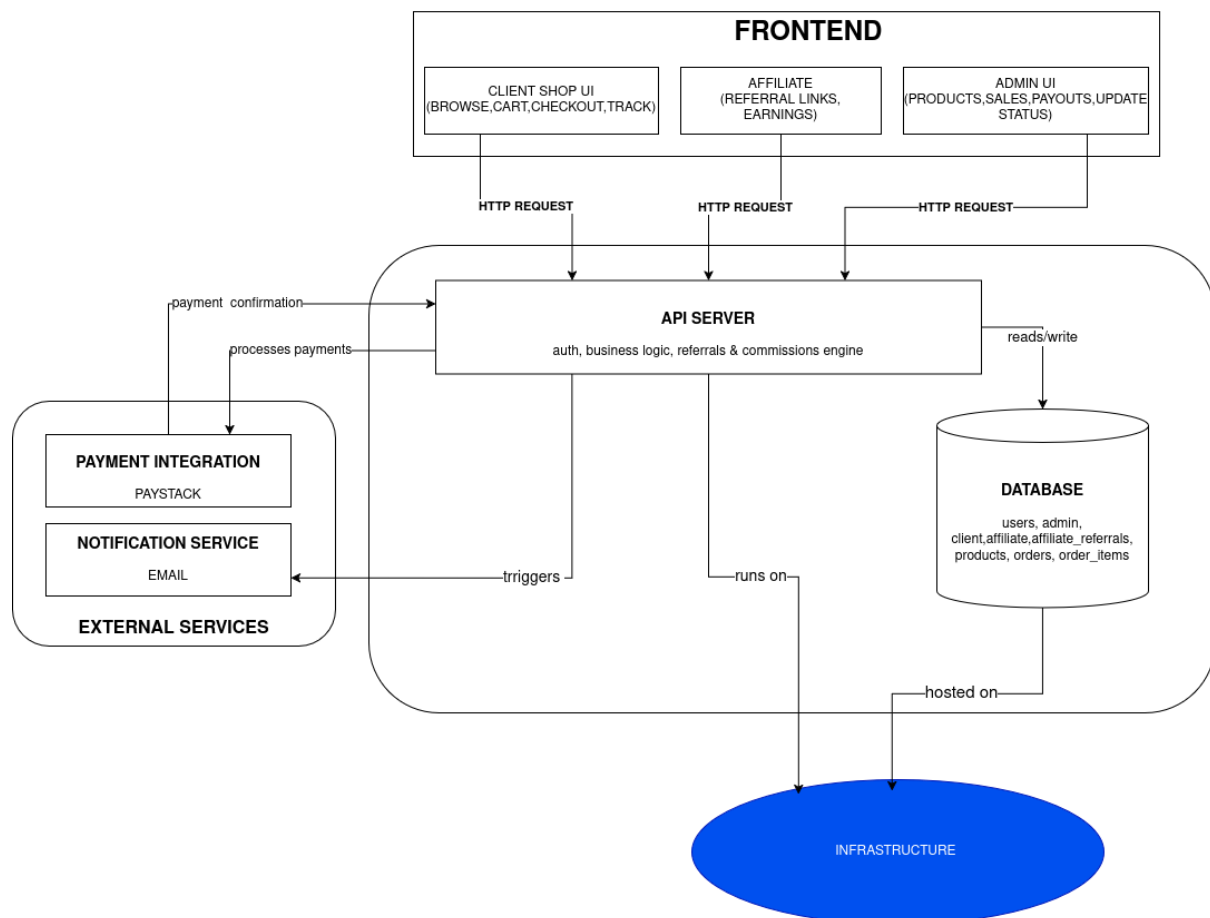
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1. Overview

PKAF LEV is an ecommerce platform with an integrated affiliate/referral system. The platform supports three user roles: Client (browses and purchases products), Affiliate/Partner (refers clients and earns commission), and Admin (manages products, sales, payouts, and affiliate accounts). This document defines the system architecture, database schema, business rules, and interaction flows that guide development.

2. System Architecture

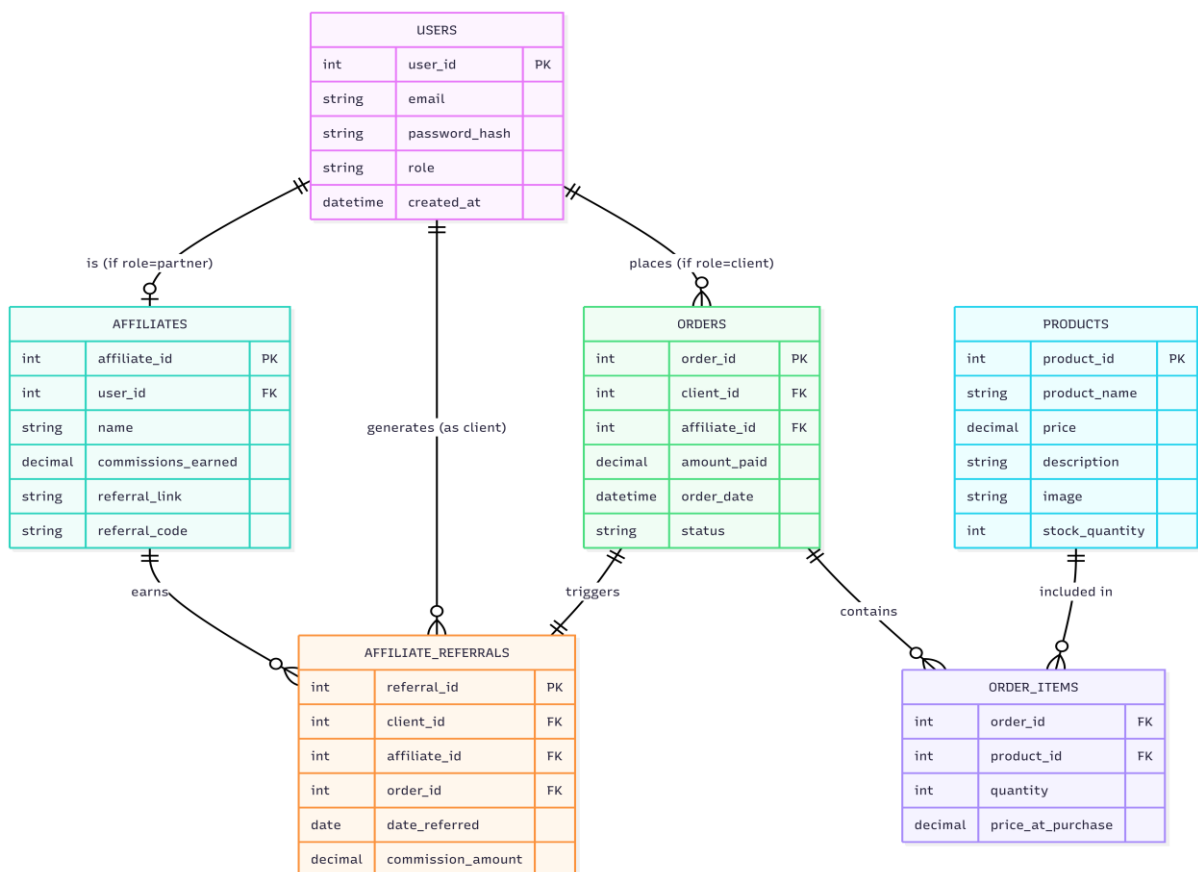
The platform consists of a frontend layer (client shop, affiliate dashboard, admin dashboard), a backend API server handling authentication and business logic, a PostgreSQL database, and external services for payment processing (Paystack/Flutterwave) and email notifications (Mailgun). All external service calls are routed through the API server; the frontend never communicates with payment or notification services directly.



3. Database Schema (Entity-Relationship Design)

The schema uses a hybrid model: a central users table holds shared authentication fields, with role-specific detail split into separate linked tables (e.g. affiliates, payouts).

Table	Key Fields
users	id, email, password_hash, role, created_at
affiliates	affiliateId, user_id, name, commissionsEarned, available_balance, number_of_referrals, referral_link, referral_code, account_status
affiliate_referrals	referral_id, clientId, affiliateId, order_id, dateReferred, commission_amount
payouts	payout_id, affiliate_id, amount_requested, status, decline_reason, date_requested, date_resolved
products	product_id, product_name, price, description, image, stock_quantity
orders	order_id, client_id, affiliate_id (optional), amount_paid, delivery_address, order_date, status
order_items	order_id, product_id, quantity, price_at_purchase

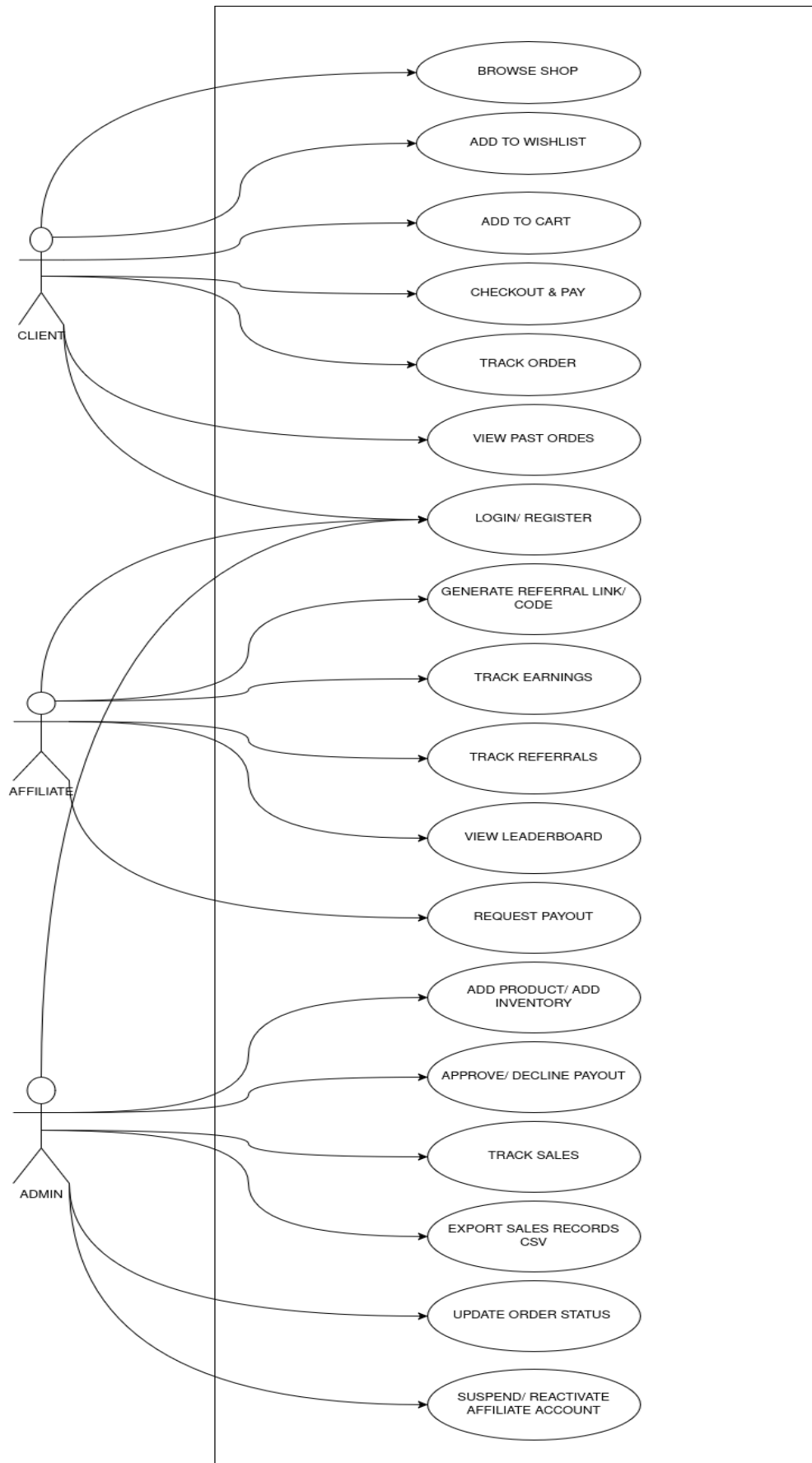


4. Business Rules

- Commission rate is a flat 3.5%, calculated on the original product price, not the discounted price.
- Referred clients (checkout with a valid referral code/link) receive a 1.5% discount on amount paid.
- Referral tracking is link-based: the referral code attaches silently in the background when a client arrives via a partner's link, no manual code entry required.
- When a referral-linked order succeeds, three affiliate fields update together inside a single database transaction: `commissionsEarned (+)`, `available_balance (+)`, `number_of_referrals (+1)`. If any step fails, all steps roll back.
- `order_items.price_at_purchase` is a permanent snapshot of price at time of sale and must never be recalculated from current product price.
- Minimum payout threshold: GHS 500. Affiliates cannot request a payout below this amount, or exceeding their current `available_balance`.
- Payout requests require Admin approval or decline. Declines must include a reason, communicated to the affiliate via email notification.
- Suspended affiliates (`account_status = suspended`): referral link/code stops attributing new referrals, payout requests are blocked, and they are excluded from the leaderboard. Already-earned commissions are retained, not erased. Suspension applies at the affiliate level only; the user retains full client access.
- Leaderboard displays affiliate first name and number of referrals only (not commission amounts), filtered to active affiliates only.
- SSL is provided via Let's Encrypt (free, browser-trusted, auto-renewed via Certbot every 90 days).
- Delivery is handled manually by the business owner; order status is manually updated by Admin (pending → out for delivery → delivered).
- Customer support handled via dedicated WhatsApp number (v1). Displayed via footer/Contact Us button on client-facing UI. Ticketing system may be considered in a future update.

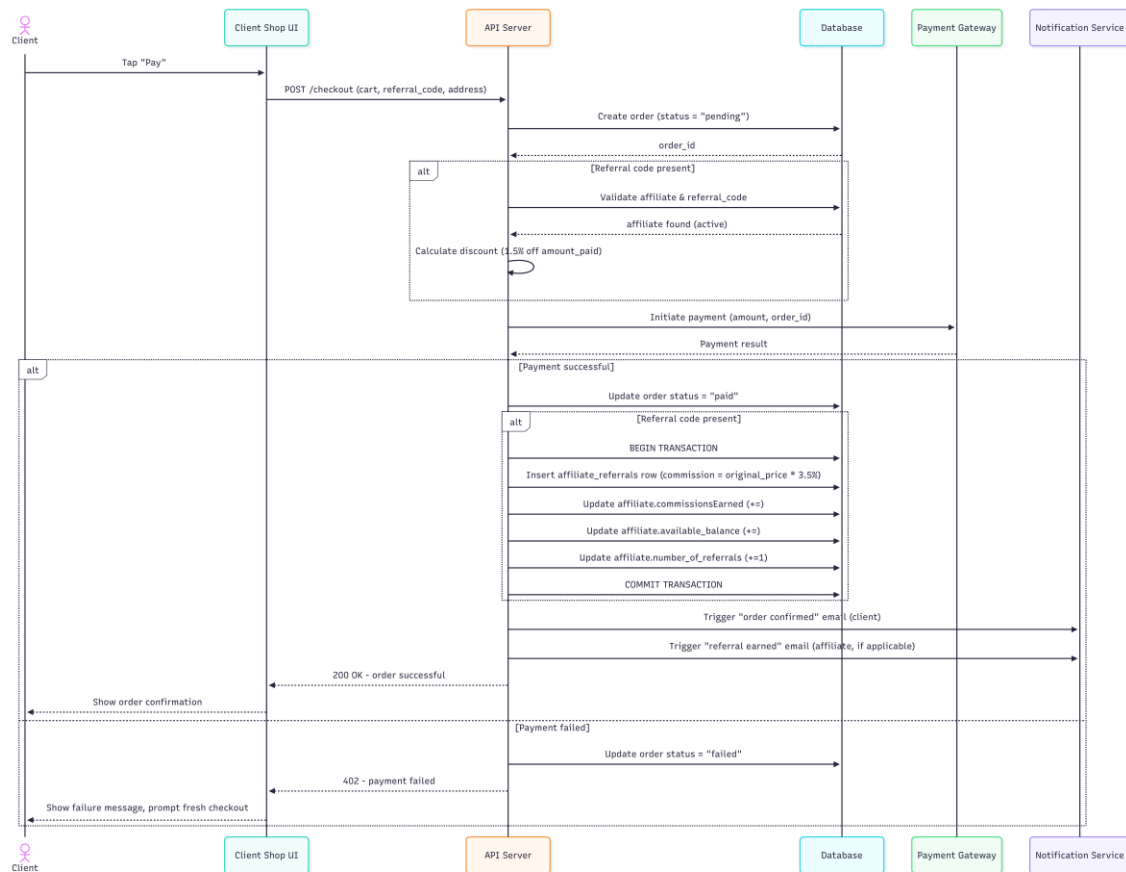
5. Use Case Diagram

Defines all actions available to each of the three actors: Client, Affiliate, and Admin.



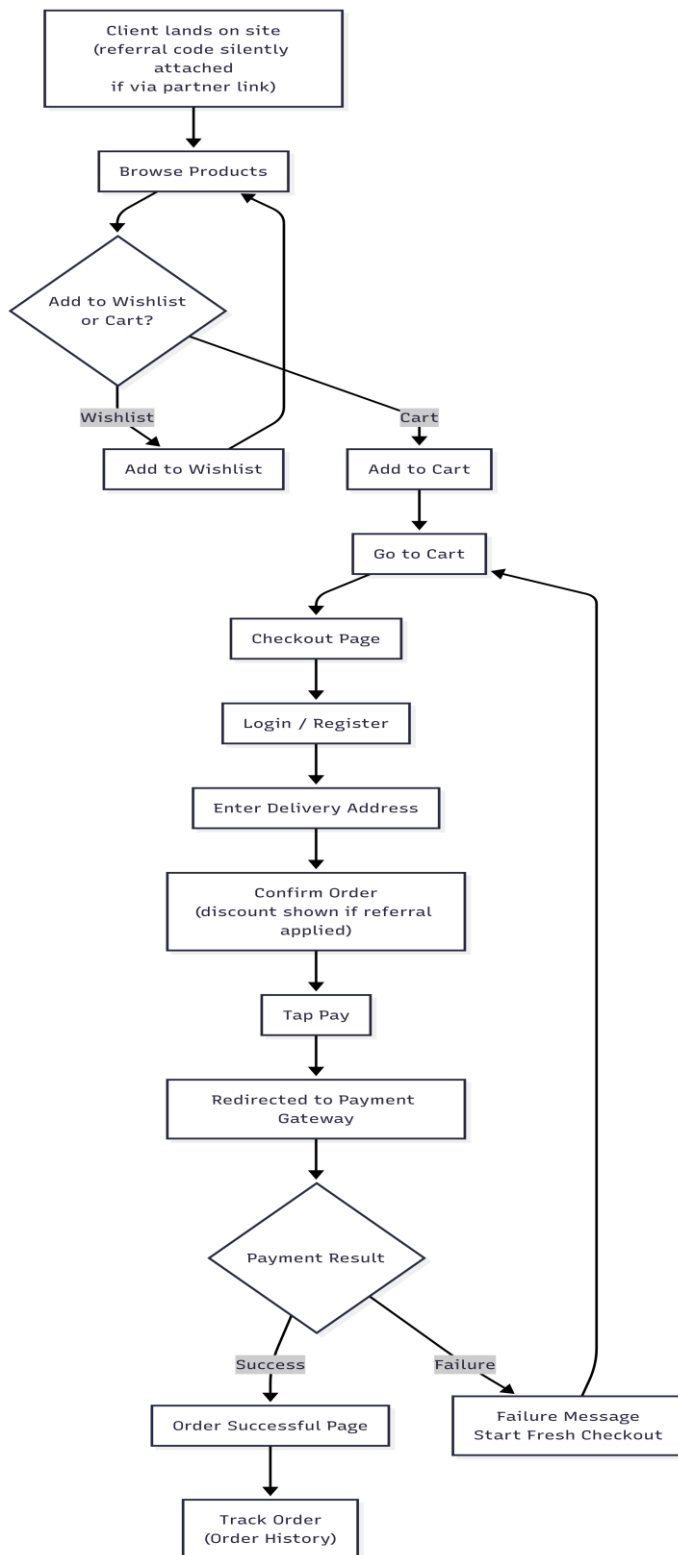
6. Sequence Diagram — Checkout & Commission Flow

Describes the full checkout process from payment initiation through order confirmation, including the referral discount calculation and the atomic commission update transaction. Failed payments leave the order permanently marked 'failed'; clients must start a fresh checkout rather than retrying the same order, avoiding duplicate-processing risk.

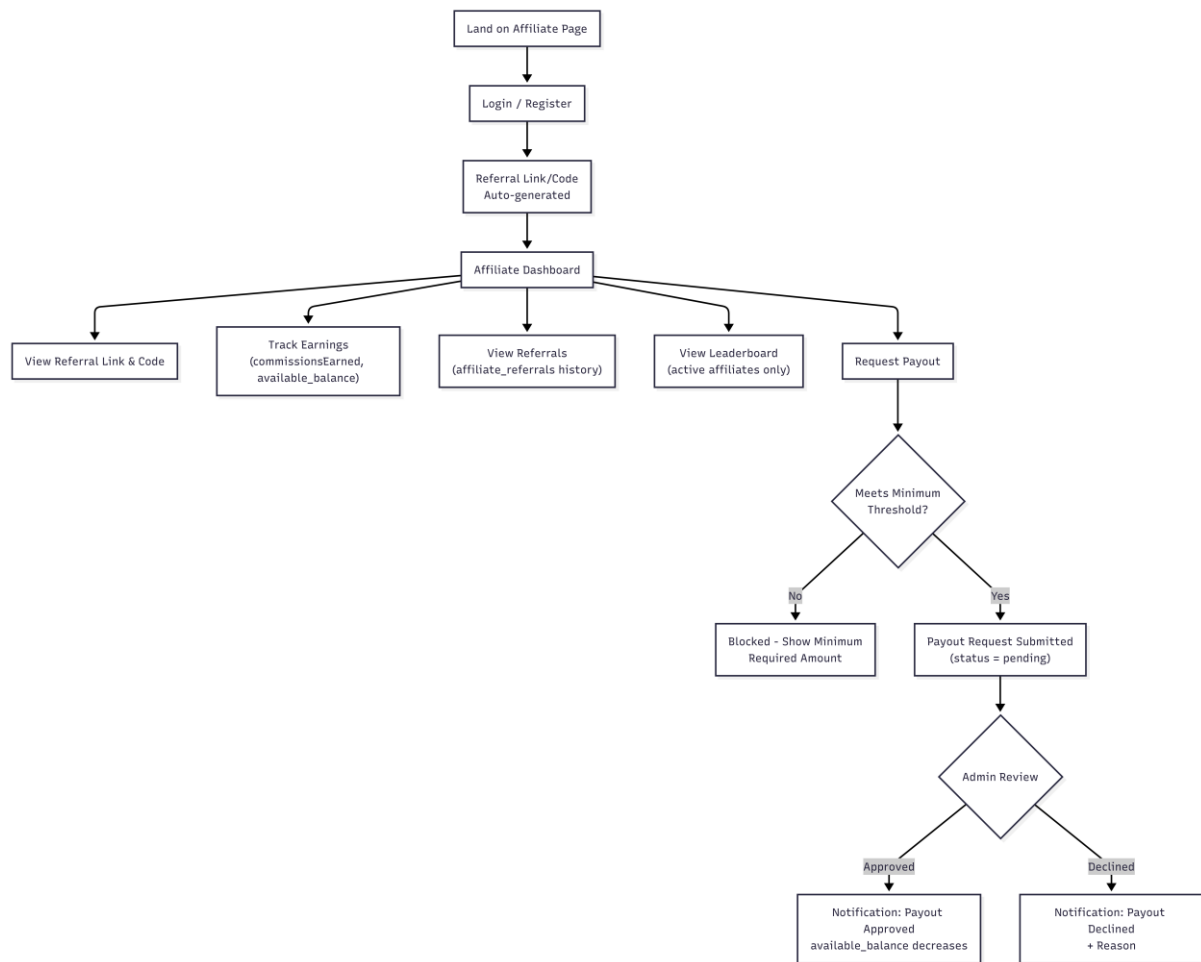


7. User Flow Diagrams

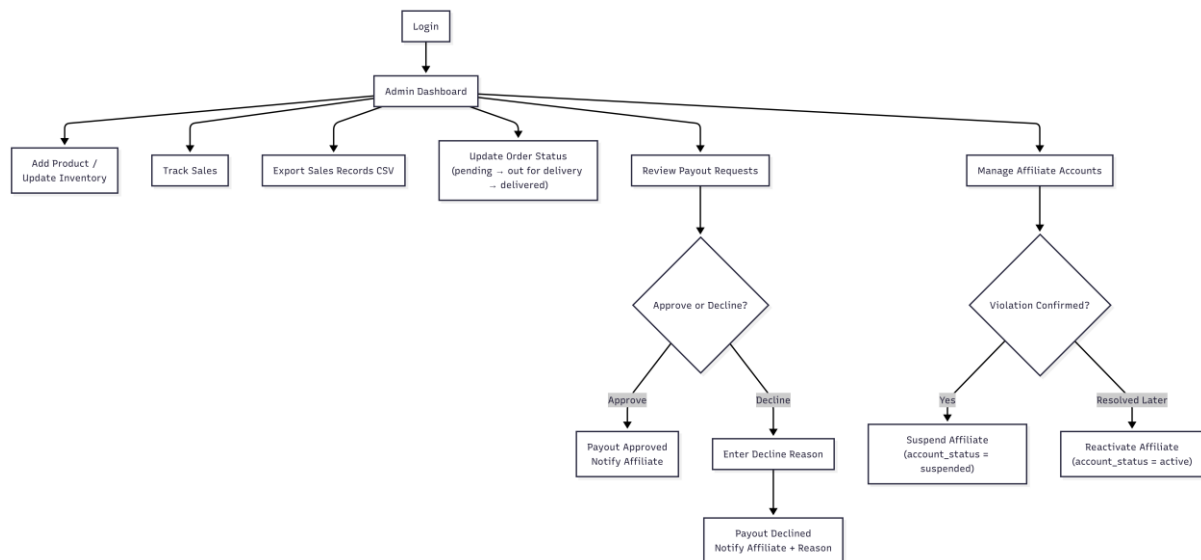
7.1 Client Flow



7.2 Affiliate Flow



7.3 Admin Flow



8. API Contract (Rough)

High-level list of actions the system must support. Request/response payload specifics to be finalized by the backend developer during implementation.

Endpoint	Purpose
POST /register	Create user account (role: client/affiliate)
POST /login	Authenticate, returns user + role + token
GET /products	List all products
GET /products/:id	Single product detail
POST /cart	Add item to cart
POST /wishlist	Add item to wishlist
POST /checkout	Create order (status: pending), returns order_id
POST /payment/initiate	Trigger payment gateway
GET /orders/:id	Track specific order
GET /orders	View order history
GET /affiliate/dashboard	Referral link/code, earnings, balance, referral count
GET /affiliate/referrals	List of affiliate_referrals rows
GET /leaderboard	Active affiliates ranked by referrals
POST /affiliate/payout-request	Submit payout request (min GHS 500)
POST /admin/products	Add product
PUT /admin/products/:id	Update product/inventory
GET /admin/sales	Sales overview
GET /admin/sales/export	CSV export of sales records
PUT /admin/orders/:id/status	Update delivery status
GET /admin/payouts	List pending payout requests
PUT /admin/payouts/:id	Approve/decline payout (reason required if declined)
PUT /admin/affiliates/:id/status	Suspend/reactivate affiliate

9. Open Questions / Future Considerations

- SMS notifications: deferred to a future update (email via Mailgun covers launch).
- Product category-based commission variation: not needed at launch (flat 3.5% rate confirmed).
- Delivery service integration: not needed at launch (business owner handles delivery manually).
- Consider whether affiliates should be allowed multiple referral links/codes in a future version (e.g. per campaign) — out of scope for v1.